Experiment: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim

To analyze the effect of steering angle variation in a MIMO beamforming system by studying

steering angle, array gain, and beam direction error using MATLAB.

Objective – 1

Analyze Steering Angle (Degrees)

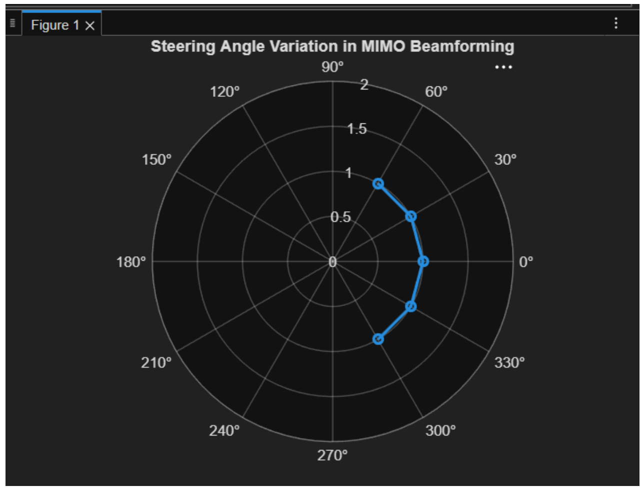
Analyze different steering angles of a MIMO beamforming system using MATLAB. Observe

how the beam can be directed toward various target directions by varying the steering angle.

MATLAB Code:

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = ones(size(angle)); % Normalized Gain

figure

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```

Expected Output



• Plot showing different steering angles.

• Beam steering visualization.

Objective – 2

Compare Array Gain (dB)

Compare the array gain obtained at different steering angles using MATLAB. Observe the

improvement in signal strength achieved through beamforming.

MATLAB Code:

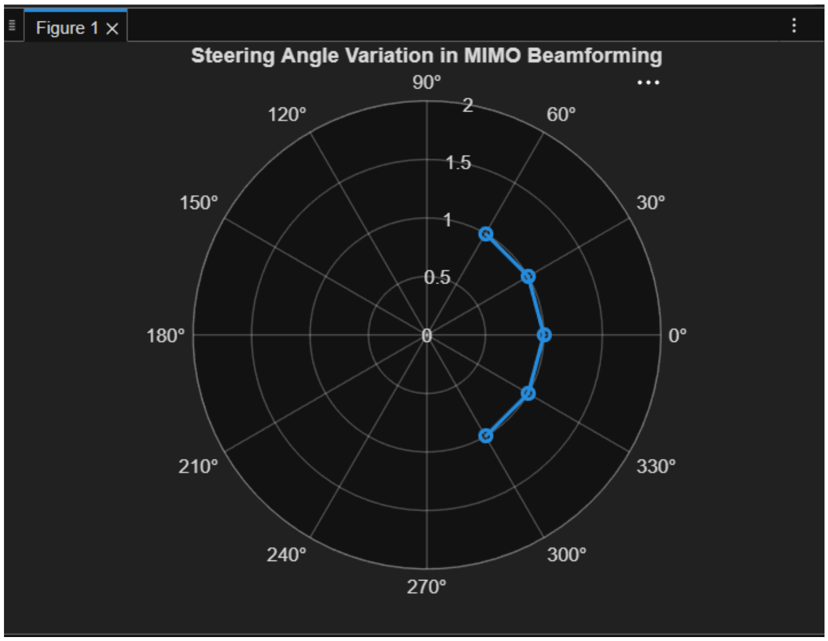
```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = [9 12 15 12 9]; % Array Gain (dB)

figure

bar(angle, gain)

grid on
```

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('Array Gain for Different Steering Angles')

Expected Output



• Bar graph showing array gain for different steering angles.

Objective – 3

Evaluate Beam Direction Error (Degrees)

Analyze the difference between the desired steering angle and the measured beam direction using

MATLAB. Compare the beam direction error for various steering angles and evaluate the beam

steering accuracy.

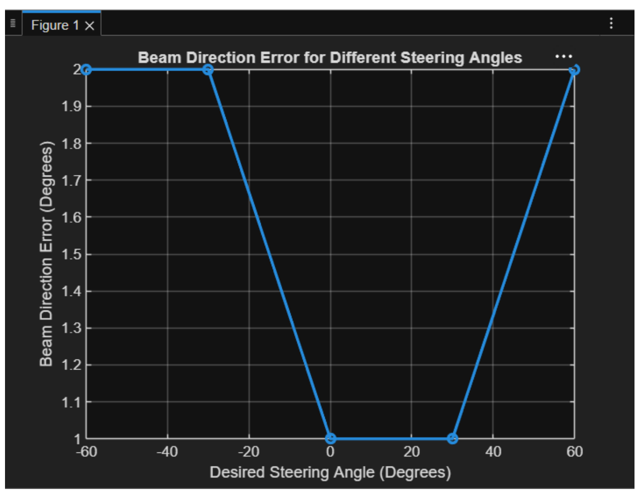
MATLAB Code:

clc;

clear;

```
close all;

desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees) actual = [-58 -
32 1

29 62]; % Actual Beam Direction (degrees)

error = abs(desired - actual); % Beam Direction Error

figure

plot(desired, error, 'o-', 'LineWidth', 2)

grid on

xlabel('Desired Steering Angle (Degrees)')

ylabel('Beam Direction Error (Degrees)')

title('Beam Direction Error for Different Steering Angles')
```

Expected Output



• Line graph showing beam direction error versus steering angle.

Result :The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.